

Ch27: 3, 9, 17, 37, 47, 69

Basic score 4` + problem 37 2`+ problem 47 2`+ problem 69 2`

3. We first find the currents. Let i_1 be the current in R_1 and take it to be positive if it is to the right. Let i_2 be the current in R_2 and take it to be positive if it is to the left. Let i_3 be the current in R_3 and take it to be positive if it is upward. The junction rule produces

$$i_1 + i_2 + i_3 = 0.$$

The loop rule applied to the left-hand loop produces

$$\mathcal{E}_1 - i_1 R_1 + i_3 R_3 = 0$$

and applied to the right-hand loop produces

$$\mathcal{E}_2 - i_2 R_2 + i_3 R_3 = 0.$$

We substitute $i_3 = -i_2 - i_1$, from the first equation, into the other two to obtain

$$\mathcal{E}_1 - i_1 R_1 - i_2 R_3 - i_1 R_3 = 0$$

$$\mathcal{E}_2 - i_2 R_2 - i_2 R_3 - i_1 R_3 = 0.$$

Solving the above equations yield

$$i_1 = \frac{\mathcal{E}_1(R_2 + R_3) - \mathcal{E}_2 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} = \frac{(1.00 \text{ V})(2.00 \Omega + 5.00 \Omega) - (3.00 \text{ V})(5.00 \Omega)}{(4.00 \Omega)(2.00 \Omega) + (4.00 \Omega)(5.00 \Omega) + (2.00 \Omega)(5.00 \Omega)} = -0.211 \text{ A}.$$

$$i_2 = \frac{\mathcal{E}_2(R_1 + R_3) - \mathcal{E}_1 R_3}{R_1 R_2 + R_1 R_3 + R_2 R_3} = \frac{(3.00 \text{ V})(4.00 \Omega + 5.00 \Omega) - (1.00 \text{ V})(5.00 \Omega)}{(4.00 \Omega)(2.00 \Omega) + (4.00 \Omega)(5.00 \Omega) + (2.00 \Omega)(5.00 \Omega)} = 0.579 \text{ A}.$$

$$i_3 = -i_1 - i_2 = -0.368 \text{ A}.$$

Note that the current i_3 in R_3 is actually downward and the current i_2 in R_2 is to the right. The current i_1 in R_1 is to the right.

(a) The power dissipated in R_1 is $P_1 = i_1^2 R_1 = (0.211 \text{ A})^2 (4.00 \Omega) = 0.178 \text{ W}.$

(b) The power dissipated in R_2 is

$$P_2 = i_2^2 R_2 = (0.579 \text{ A})^2 (2.00 \, \Omega) = 0.670 \text{ W}.$$

(c) The power dissipated in R_3 is $P_3 = i_3^2 R_3 = (-0.368 \text{ A})^2 (5.00 \, \Omega) = 0.677 \text{ W}.$

(d) The power supplied by $\mathcal{E}_1$ is $i_1 \mathcal{E}_1 = (-0.211 \text{ A})(1.00 \text{ V}) = -0.211 \text{ W}.$ The negative sign indicates that $\mathcal{E}_1$ is actually absorbing energy from the circuit.

(e) The power “supplied” by $\mathcal{E}_2$ is $i_2 \mathcal{E}_2 = (0.579 \text{ A})(3.00 \text{ V}) = 1.737 \text{ W}.$

9. (a) In the steady state situation, the capacitor voltage will equal the voltage across $R_2 = 15 \text{ k}\Omega$:

$$V_0 = R_2 \frac{\mathcal{E}}{R_1 + R_2} = (15.0 \text{ k}\Omega) \left(\frac{20.0 \text{ V}}{10.0 \text{ k}\Omega + 15.0 \text{ k}\Omega} \right) = 12.0 \text{ V}.$$

Now, multiplying $q = q_0 e^{-t/RC}$ by the capacitance leads to $V = V_0 e^{-t/RC}$ describing the voltage across the capacitor (and across $R_2 = 15.0 \text{ k}\Omega$) after the switch is opened (at $t = 0$). Thus, with $t = 0.00400 \text{ s}$, we obtain

$$V = (12) e^{-0.004 / (15000)(0.4 \times 10^{-6})} = 6.16 \text{ V}.$$

Therefore, using Ohm’s law, the current through R_2 is $6.16 / 15000 = 4.11 \times 10^{-4} \text{ A}.$

$$(b) \frac{di}{dt} = -\frac{V_0}{R} e^{-t/RC} \left(\frac{1}{RC} \right) = -68.5 \text{ mA} / \text{s}$$

$$(c) \frac{dP}{dt} = \frac{d(i^2 R)}{dt} = 2iR \frac{di}{dt} = -0.84 \text{ W} / \text{s}$$

17. The currents in R_1, R_2, R_3 are i_1, i_2 and i_3 , then:

$$i_1 = \mathcal{E} / (R_1 + R_{23})$$

$$i_1 = i_2 + i_3$$

$$i_2 R_2 = i_3 R_3$$

Thus,

$$i_3 = \frac{R_2}{R_2 + R_3} \frac{\varepsilon}{R_1 + R_{23}} = \frac{R_2 \varepsilon}{(R_2 + R_3)(R_1 + \frac{R_2 R_3}{R_2 + R_3})} = \frac{R_2 \varepsilon}{(R_1 R_2 + R_1 R_3 + R_2 R_3)}$$

$$P_3 = i_3^2 R_3 = (R_2 \varepsilon)^2 R_3 (R_1 R_2 + R_1 R_3 + R_2 R_3)^{-2} = 49 \varepsilon^2 R_3 (21 + 10 R_3)^{-2} = 49 \varepsilon^2 f(R_3)$$

where we use the equivalence symbol $\equiv$ to define the expression $f(R_3)$. To maximize P_3 we need to minimize the expression $f(R_3)$. We set

$$\frac{df(R_3)}{dR_3} = 0 \Rightarrow (21 + 10 R_3)^{-2} - 20 R_3 (21 + 10 R_3)^{-3} = 0 \Rightarrow R_3 = 2.1 \Omega$$

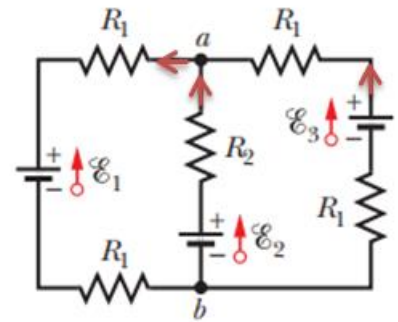
37. (a) We note that the R_1 resistors occur in series pairs, contributing net resistance $2R_1$ in each branch where they appear. Since $\varepsilon_2 = \varepsilon_3$ and $R_2 = 2R_1$, from symmetry we know that the currents through ε_2 and ε_3 are the same: $i_2 = i_3 = i$. Therefore, from the junction rule we have the current through ε_1 is

$$i_1 = i_2 + i_3 = 2i.$$

Apply loop rule for the left side loop, we have:

$$\varepsilon_2 - iR_2 - (2R_1)(2i) - \varepsilon_1 = 0 \quad \text{0.5 point}$$

$$i = \frac{\varepsilon_2 - \varepsilon_1}{4R_1 + R_2} = \frac{4.0 \text{ V} - 2.0 \text{ V}}{4(1.0 \Omega) + 2.0 \Omega} = 0.33 \text{ A}.$$



Therefore, the current through ε_1 is $i_1 = 2i = 0.67 \text{ A}$. **0.2 point**

(b) The direction of i_1 is downward. **0.2 point**

(c) The current through ε_2 is $i_2 = 0.33 \text{ A}$. **0.1 point**

(d) The direction of i_2 is upward. **0.1 point**

(e) Battery 1 is absorbing energy. **0.2 point**

(f) $P = \varepsilon_1 i_1 = 1.34 \text{ W}$. **0.2 point**

(g) $V_a - V_b = -iR_2 + \varepsilon_2 = -(0.333 \text{ A})(2.0 \Omega) + 4.0 \text{ V} = 3.3 \text{ V}$.

0.5 point (公式 0.3 分, 结果 0.2 分)

47. At $t = 0$ the capacitor is completely uncharged and the current in the capacitor branch is as it would be if the capacitor were replaced by a wire. Let i_1 be the current in R_1 and take it to be positive if it is to the right. Let i_2 be the current in R_2 and take it to be positive if it is downward. Let i_3 be the current in R_3 and take it to be positive if it is downward. The junction rule produces $i_1 = i_2 + i_3$, the loop rule applied to the left-hand loop produces

$$\varepsilon - i_1 R_1 - i_2 R_2 = 0 ,$$

and the loop rule applied to the right-hand loop produces

$$i_2 R_2 - i_3 R_3 = 0 .$$

Since the resistances are all the same we can simplify the mathematics by replacing R_1 , R_2 , and R_3 with R .

(a) Solving the three simultaneous equations, we find

$$i_1 = \frac{2\varepsilon}{3R} = \frac{2 \times 1.2 \times 10^3 \text{ V}}{3(0.73 \times 10^6 \Omega)} = 1.09 \times 10^{-3} \text{ A}, \quad \text{0.3 point}$$

$$(b) \quad i_2 = \frac{\varepsilon}{3R} = \frac{1.2 \times 10^3 \text{ V}}{3(0.73 \times 10^6 \Omega)} = 5.5 \times 10^{-4} \text{ A}, \quad \text{0.3 point}$$

$$(c) \quad i_3 = i_2 = 5.5 \times 10^{-4} \text{ A}. \quad \text{0.2 point}$$

At $t = \infty$ the capacitor is fully charged and the current in the capacitor branch is 0. Thus, $i_1 = i_2$, and the loop rule yields

$$\varepsilon - i_1 R_1 - i_1 R_2 = 0 .$$

(d) The solution is

$$i_1 = \frac{\varepsilon}{2R} = \frac{1.2 \times 10^3 \text{ V}}{2(0.73 \times 10^6 \Omega)} = 8.2 \times 10^{-4} \text{ A}. \quad \text{0.3 point}$$

$$(e) \quad i_2 = i_1 = 8.2 \times 10^{-4} \text{ A}. \quad \text{0.2 point}$$

(f) As stated before, the current in the capacitor branch is $i_3 = 0$. 0.2 point

(g) For $t=0$, there is no charge on the capacitor, so the potential difference across the capacitor is 0, so the capacitor looks like a wire and R_2 and R_3 are in parallel. So we have:

$$V_2 = \varepsilon/3 = (1.2 \times 10^3 \text{ V})/3 = 4.0 \times 10^2 \text{ V}. \quad 0.3 \text{ point}$$

(h) For $t = \infty$, the charge on the capacitor does not change, so there is no current in R_3 and capacitor. R_1 and R_2 just in series with battery, so we have:

$$V_2 = \varepsilon/2 = (1.2 \times 10^3 \text{ V})/2 = 6.0 \times 10^2 \text{ V}. \quad 0.2 \text{ point}$$

(i) We take the upper plate of the capacitor to be positive. This is consistent with current flowing into that plate. The junction equation is $i_1 = i_2 + i_3$, and the loop equations are

$$\begin{aligned} \varepsilon - i_1 R - i_2 R &= 0 \\ -\frac{q}{C} - i_3 R + i_2 R &= 0. \end{aligned}$$

We use the first equation to substitute for i_1 in the second and obtain $\varepsilon - 2i_2 R - i_3 R = 0$. Thus $i_2 = (\varepsilon - i_3 R)/2R$. We substitute this expression into the third equation above to obtain

$$-(q/C) - (i_3 R) + (\varepsilon/2) - (i_3 R/2) = 0.$$

Now we replace i_3 with dq/dt to obtain

$$\frac{3R}{2} \frac{dq}{dt} + \frac{q}{C} = \frac{\varepsilon}{2}.$$

This is just like the equation for an RC series circuit, except that the time constant is $\tau = 3RC/2$ and the impressed potential difference is $\varepsilon/2$. The solution is

$$q = \frac{C\varepsilon}{2} (1 - e^{-2t/3RC})$$

The current in the capacitor branch is

$$i_3(t) = \frac{dq}{dt} = \frac{\mathcal{E}}{3R} e^{-2t/3RC}.$$

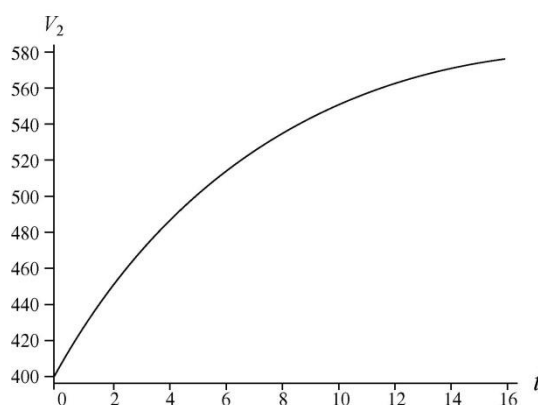
The current in the center branch is

$$i_2(t) = \frac{\mathcal{E}}{2R} - \frac{i_3}{2} = \frac{\mathcal{E}}{2R} - \frac{\mathcal{E}}{6R} e^{-2t/3RC} = \frac{\mathcal{E}}{6R} (3 - e^{-2t/3RC})$$

and the potential difference across R_2 is

$$V_2(t) = i_2 R = \frac{\mathcal{E}}{6} (3 - e^{-2t/3RC}).$$

So A plot of V_2 as a function of time is shown in the following graph.



69. Let i_1 be the current in R_1 and take it to be positive if it is to the right. Let i_2 be the current in R_2 and take it to be positive if it is upward.

(a) When the loop rule is applied to the lower loop, the result is

$$\mathcal{E}_2 - i_1 R_1 = 0.$$

The equation yields

$$i_1 = \frac{\mathcal{E}_2}{R_1} = \frac{10.0 \text{ V}}{100\Omega} = 0.10 \text{ A. } \text{0.5 point (公式 0.3 分, 结果 0.2 分)}$$

(b) When it is applied to the upper loop, the result is

$$\mathcal{E}_1 - \mathcal{E}_2 - \mathcal{E}_3 - i_2 R_2 = 0 . \quad 0.5 \text{ point}$$

The equation gives

$$i_2 = \frac{\mathcal{E}_1 - \mathcal{E}_2 - \mathcal{E}_3}{R_2} = \frac{6.0 \text{ V} - 10.0 \text{ V} - 4.0 \text{ V}}{50 \Omega} = -0.160 \text{ A} , \quad 0.5 \text{ point}$$

or $|i_2| = 0.160 \text{ A}$. The negative sign indicates that the current in R_2 is actually downward.

(c) If V_b is the potential at point b , then the potential at point a is $V_a = V_b + \mathcal{E}_3 + \mathcal{E}_2$, so

$$V_a - V_b = \mathcal{E}_3 + \mathcal{E}_2 = 4.0 \text{ V} + 10.0 \text{ V} = 14.0 \text{ V} . \quad 0.5 \text{ point} \quad (\text{公式 } 0.3 \text{ 分} , \text{ 结果 } 0.2 \text{ 分})$$